



PROJECT

ROTATING EQUIPMENT MAINTENANCE AND SERVICES – PACKAGE 2 No. SPHR01230A

WO NO.

8400211119

SPK NO.

1

TITLE

VIBRATION REPORT

JOB DESCRIPTION

Location

MINAS GS-1

Equipment

SB 1.3

Inspect Date

08/04/2026

Prepared,

Reviewed,

Approved,

M Azmal Fahri

Winarko

Purnama

Zul Fadhlán

Pump Specialist

Asst. Site Managaer

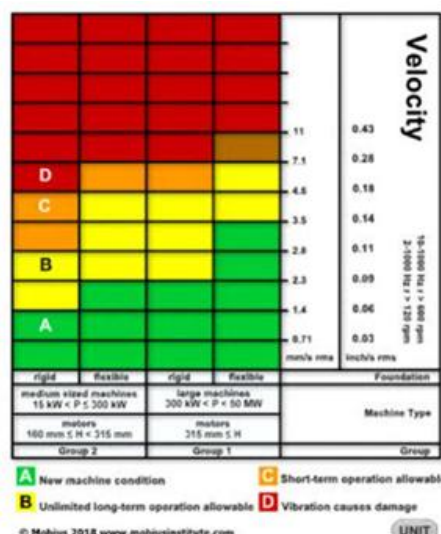
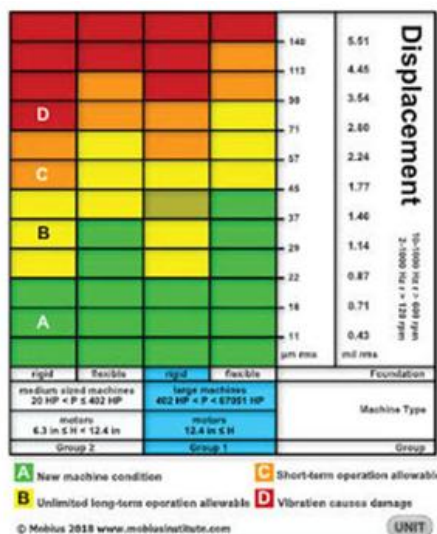
Site Manager

Spv Maintenance CS
South

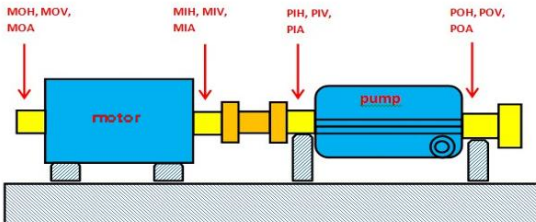
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MACHINE CLASS : CLASS 3 BASED ON ISO 10816-3		PUMP DATA	
Zone A:	Vibration of newly commissioned machine (Normal).	SERIAL NO.	: -
Zone B:	Machines considered acceptable for unrestricted long-term operation (Normal).	TAG NO.	: SB 1.3
Zone C:	Machines considered unsatisfactory for long-term continuous operation (Alert).	SERVICE	: WATER INJECTION
Zone D:	Vibration values normally considered to be sufficient severity to cause damage to the machine (Fault).	SIZE	: -
		CAPACITY	: 3000 USGPM
		TDH	: 1754 ft
		POWER	: 1750 HP
		NPSHR	: - ft
		SPEED	: 3561 RPM

ALARM LIMIT	NORMAL	ALERT	FAULT	RECENT STATUS
Overall Velocity (in/sec)	≤ 0.18	0.19-0.27	≥ 0.28	NORMAL
Overall Displacement (mils-pp)	≤ 2.9	3.0-3.9	≥ 4	NORMAL
TEMP BEARING (°C)	≤ 80	81-86	≥ 87	NORMAL



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OPERATING PARAMENTER				MACHINE CONFIGURATION
Highest Overall Vel	:	0.13	in/s RMS (MIV)	
Highest Overall Disp	:	1.7	mils-pp (MIX)	
Suction Pressure	:	8	Psi	
Discharge Pressure	:	761	Psi	
Bearing Temp M-Out	:	82	°C	
Bearing Temp M-In	:	81	°C	
Bearing Temp P-In	:	69	°C	
Bearing Temp P-Out	:	64	°C	
Flow rate	:	-	USGPM	

Finding
Overall vibration in the motor has increased, with a rise at the MIV point. The pump's spectrum is dominated by 1x TS, accompanied by harmonics up to 5x TS.
Analysis
Spectrum indicates a mechanical issue related to the motor-pump coupling system, most likely related to coupling misalignment. However: 1. Pipe strain 2. Motor soft foot 3. Insufficient hold-down preload should be verified
Recommendation
1. Confirm motor soft foot condition 2. Confirm hold-down bolt torque against the specified value 3. Confirm pipe strain on suction and discharge piping 4. Perform shaft alignment if any mechanical deviation is found 5. Re-measure vibration after action to confirm improvement

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OVERALL LAST MEASUREMENT SUMMARY

Measurement Point	OVERALL LEVEL
MOH - Motor Outboard Horizontal	0.05 in/s RMS
MOV - Motor Outboard Vertical	0.06 in/s RMS
MOA - Motor Outboard Axial	0.10 in/s RMS
MIH - Motor Inboard Horizontal	0.04 in/s RMS
MIV - Motor Inboard Vertical	0.13 in/s RMS
MIA - Motor Inboard Axial	0.12 in/s RMS
PIH - Pump Inboard Horizontal	0.08 in/s RMS
PIV - Pump Inboard Vertical	0.03 in/s RMS
PIA - Pump Inboard Axial	0.03 in/s RMS
POH - Pump Outboard Horizontal	0.06 in/s RMS
POV - Pump Outboard Vertical	0.08 in/s RMS
POA - Pump Outboard Axial	0.04 in/s RMS
MOX - Motor Outboard X	0.9 mils-pp
MOY - Motor Outboard Y	1.3 mils-pp
MIX - Motor Inboard X	1.7 mils-pp
MIY - Motor Inboard Y	1.2 mils-pp
PIX - Pump Inboard X	0.4 mils-pp
PIY - Pump Inboard Y	0.4 mils-pp
POX - Pump Outboard X	1.0 mils-pp
POY - Pump Outboard Y	0.8 mils-pp



VIBRATION REPORT



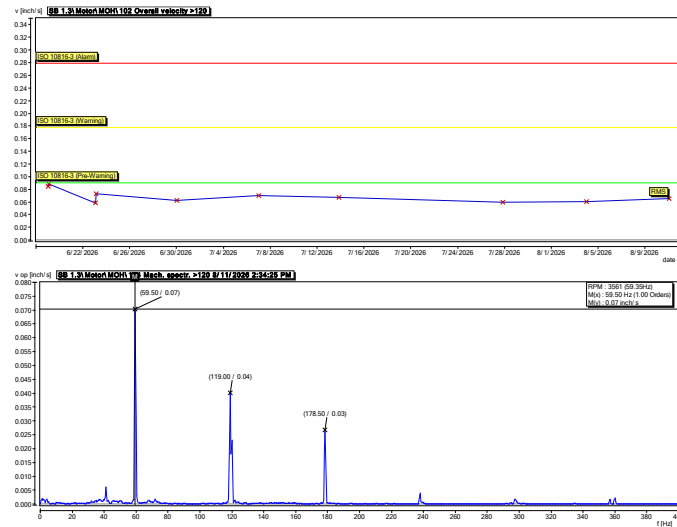
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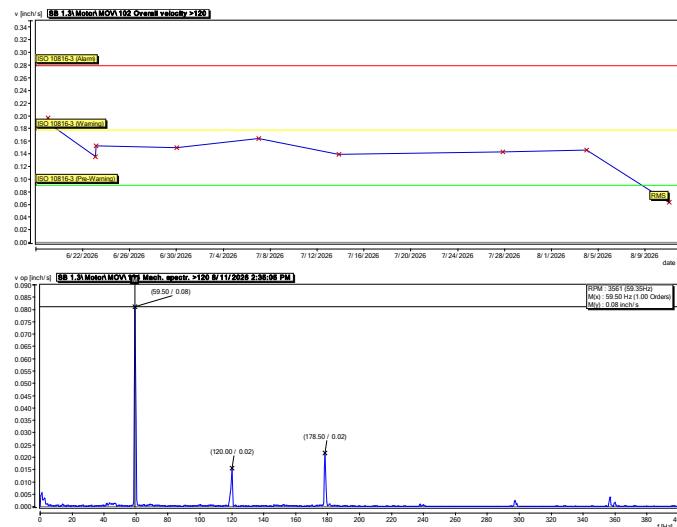
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TREND AND SPECTRUM

MOH - Motor Outboard Horizontal



MOV - Motor Outboard Vertical





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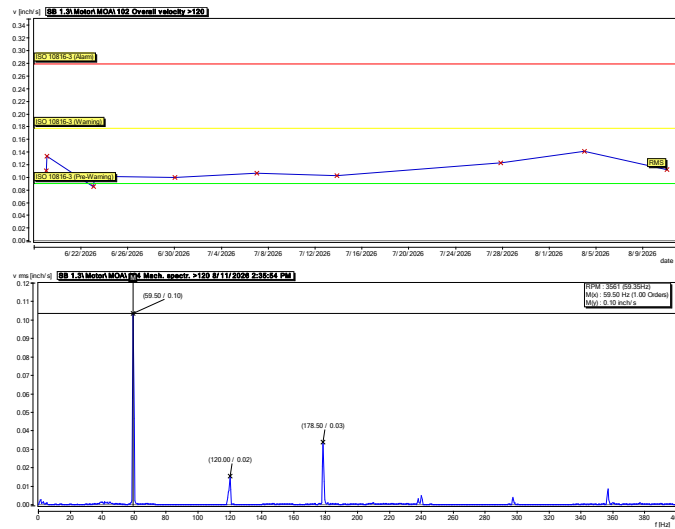


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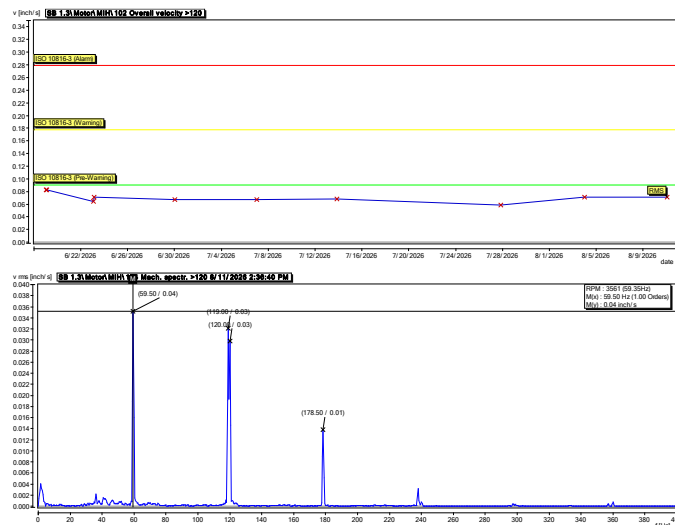
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MOA - Motor Outboard Axial



MIH - Motor Inboard Horizontal





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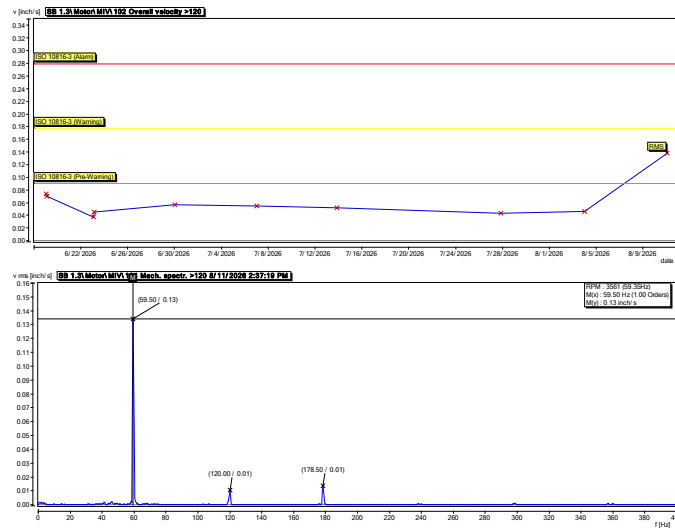


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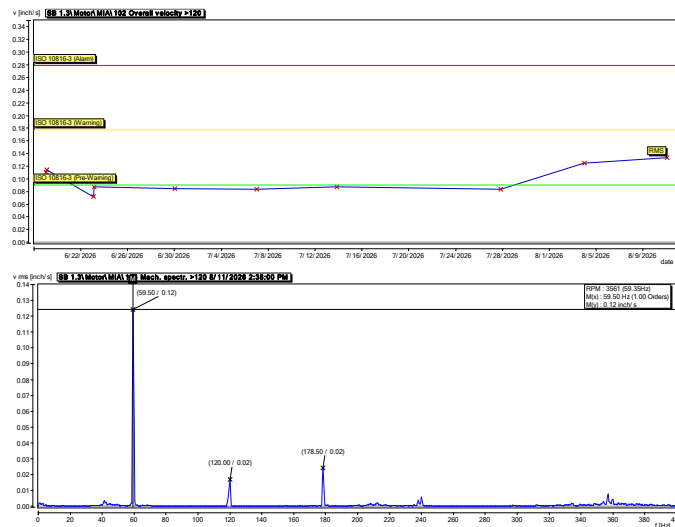
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MIV - Motor Inboard Vertical



MIA - Motor Inboard Axial





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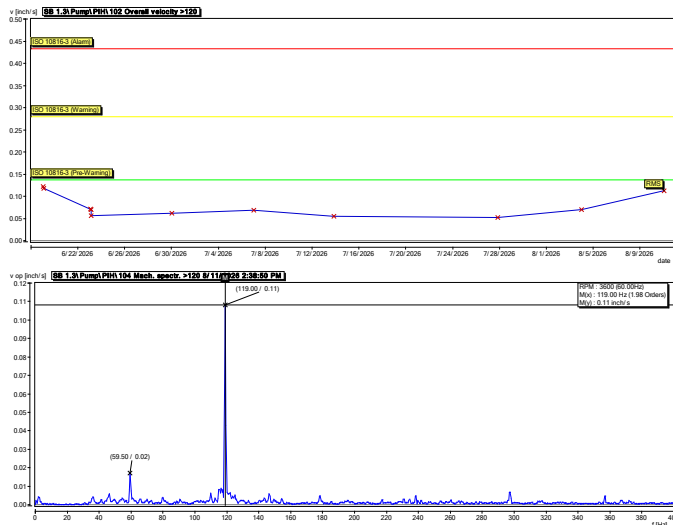


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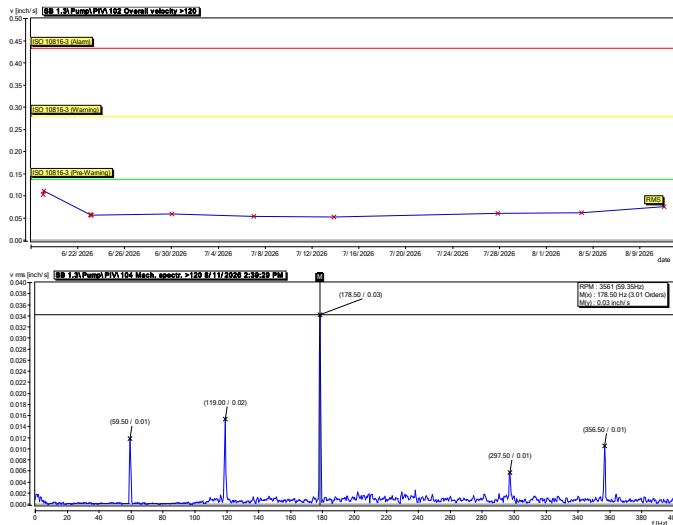
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PIH - Pump Inboard Horizontal



PIV - Pump Inboard Vertical





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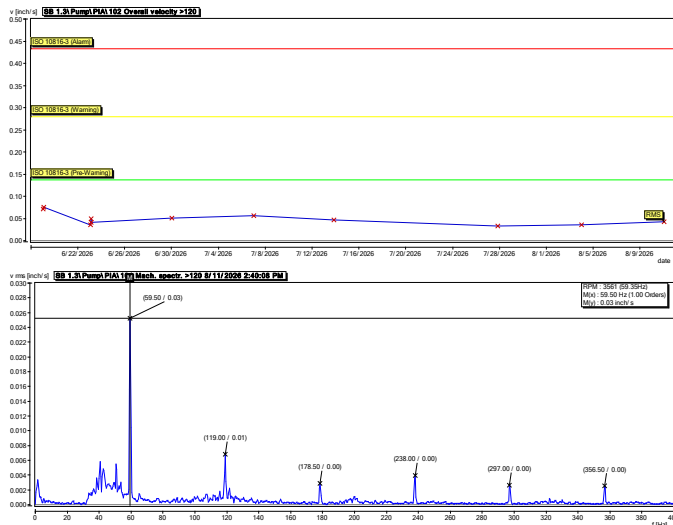


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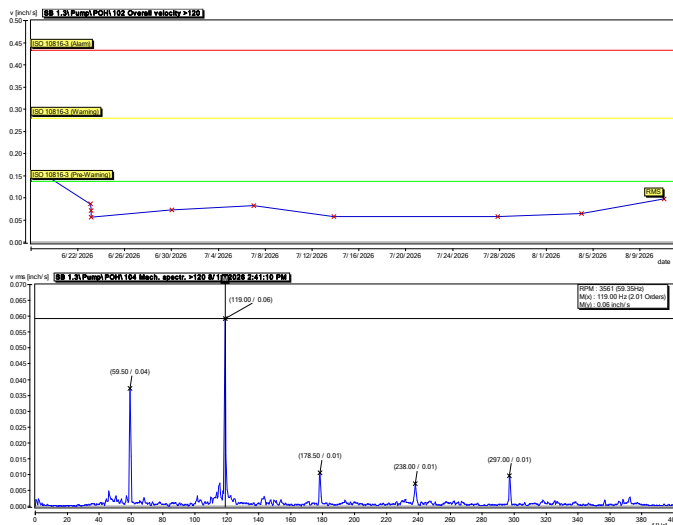
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PIA - Pump Inboard Axial



POH - Pump Outboard Horizontal





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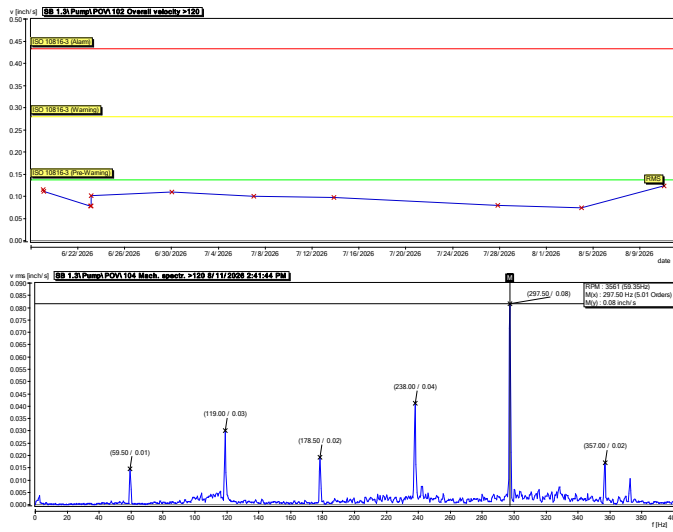


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POV - Pump Outboard Vertical



POA - Pump Outboard Axial

